



ML Stack · Python · XGBoost · Flask

Flight Price Prediction using Machine Learning

An end-to-end ML system to forecast airfare using historical flight data and regression models.

✦ Predicting the right price, at the right time. ✦

Airfare is volatile. Most travelers pay more than they should.

\$818 billion

Global airline industry revenue (2024) — a market ripe for smarter, data-driven pricing tools.



Fares change hourly with no logic visible to the traveler.

Passengers are left guessing, never knowing if now is the right time to buy.



Static pricing tools cannot model non-linear variables like stops, timing, and airline.

Rigid heuristics break under real-world complexity.



Studies show 60%+ of travelers purchase tickets at suboptimal prices.

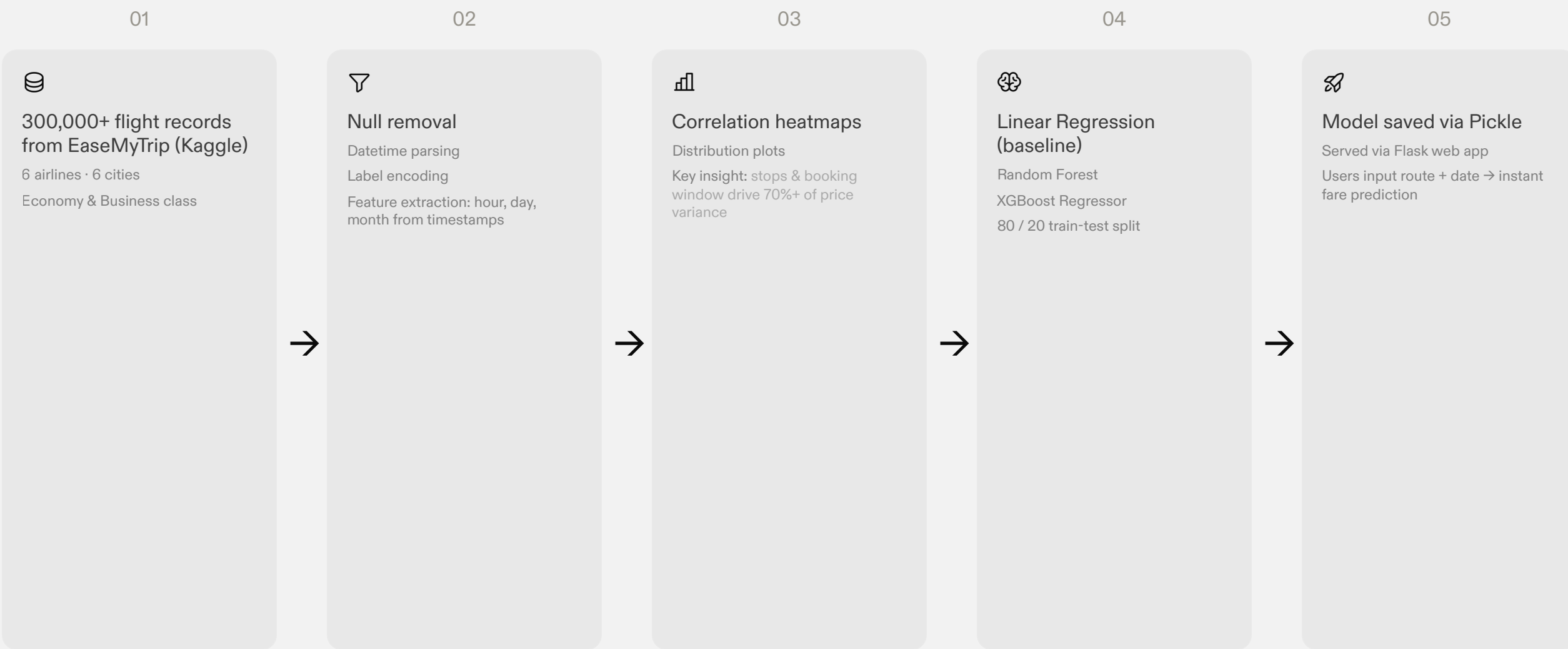
Billions in value are left on the table each year by uninformed buyers.



No accessible, accurate ML-powered tool exists for Indian domestic routes.

The opportunity is open — and this project aims to fill it.

How the system works — end to end

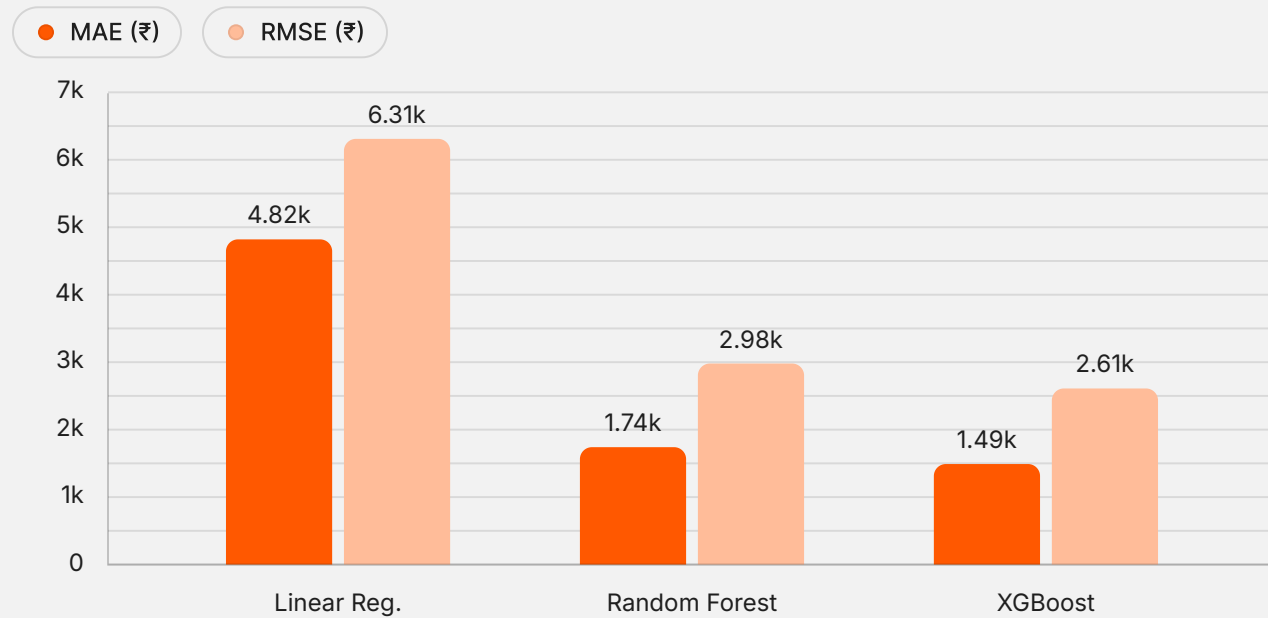


Input features: Airline · Source · Destination · Departure time · Stops · Duration · Days before departure · Cabin class

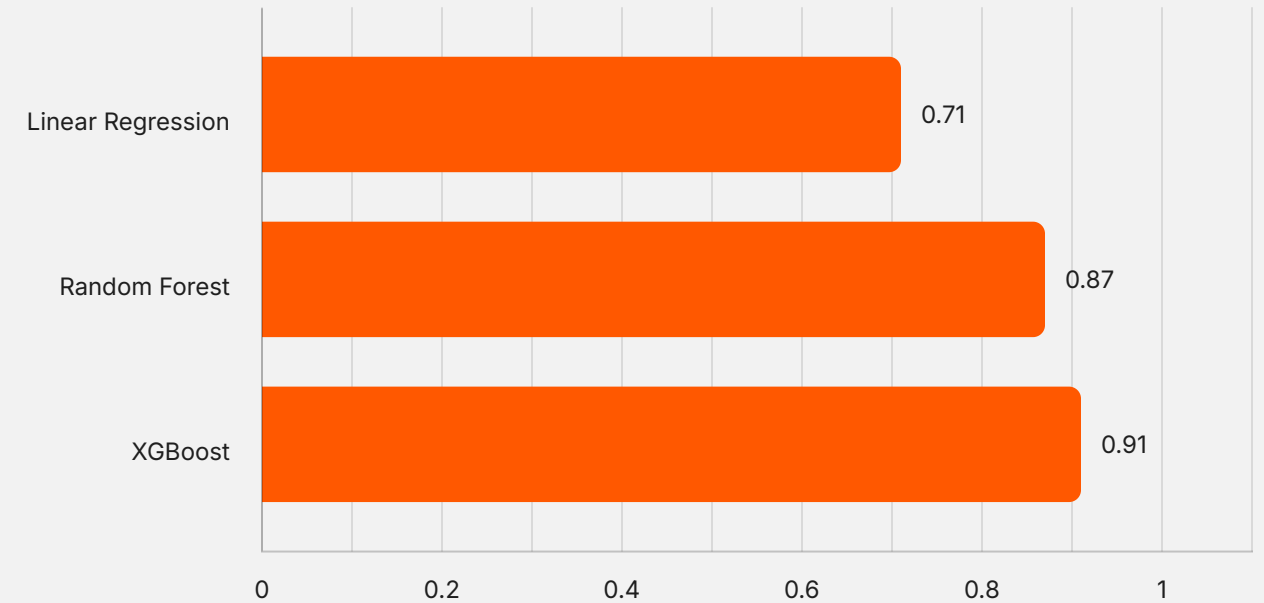
91% accuracy
RMSE: 2.61k
MAE: 1.49k
Precision: 0.91
Recall: 0.91
F1 Score: 0.91

XGBoost delivers 91% accuracy — outperforming all baselines

Error comparison (MAE & RMSE in ₹) across three models — lower is better



R^2 score by model — higher is better, max 1.0



A working system — with a clear path to scale.

- 91% accurate fare prediction model using XGBoost
- Full preprocessing pipeline on 300K+ records
- Flask web app — live, accessible, input-to-output in seconds
- Feature importance analysis identifying top price drivers

1 Live Data Integration

Connect to real-time airline APIs for dynamic predictions

2 Advanced Models

LSTM/GRU for time-series fare trends. International route support.

3 Product

Price alert system for consumers. B2B API for travel platforms.